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The tectonic and geomagnetic significance of paleomagnetic observations from volcanic rocks from central Afar, Africa

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Abstract

Deformation throughout Afar over the past 2 myr has been characterized by widespread and intense crustal fragmentation that results from inhomogeneous extension across the region. In eastern Afar, this situation has evolved to localized extension associated with the westward propagation of the Gulf of Aden/Gulf of Tadjurah seafloor spreading system into the Asal–Ghoubbet Rift. During the gradual process of rift propagation and localization, crustal blocks in eastern Afar sustained clockwise rotations of $\sim 11°$. To better understand the processes of rift propagation and localization and how they affect the rest of Afar, we have collected and analyzed over 400 oriented paleomagnetic samples from 67 lava flows from central and southern Afar. Unlike eastern Afar, the mean paleomagnetic direction from central Afar indicates that vertical-axis rotations are statistically insignificant ($3.6° \pm 4.4°$), though small clockwise rotations ($< 8°$) are permitted. Thus, propagation and localization in central Afar have not had the same influence in causing crustal block rotations or, perhaps more likely, have not reached the same stage of evolution as seen in eastern Afar. In addition, several of the lava flows record intriguing geomagnetic field behavior associated with polarity transitions, excursions, or large secular variation events. Interestingly, the transitional or anomalous virtual geomagnetic poles (VGPs) tend to cluster in two nearly antipodal regions, one in the northern Pacific Ocean and the other in the southwest Indian Ocean. One lava flow has recorded both of the antipodal transitional components, with the two components residing in magnetic minerals with unblocking temperatures above and below $\sim 500°C$, respectively. Reheating and partial remagnetization by the overlying flow cannot explain either of the transitional directions because both differ significantly from that of the reversely magnetized overlying flow. The high-temperature component gives a VGP in the northern Pacific, whereas the lower-temperature component gives a nearly antipodal VGP south of Cape Town, South Africa. Hence, the configuration of the geomagnetic field appears to have jumped nearly instantaneously from a northern-hemisphere transitional state to a southern-hemisphere one during this normal-to-reverse polarity transition. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; rifting; tectonics; magnetic field; reversals

1. Introduction

As the Gulf of Aden spreading center propa-

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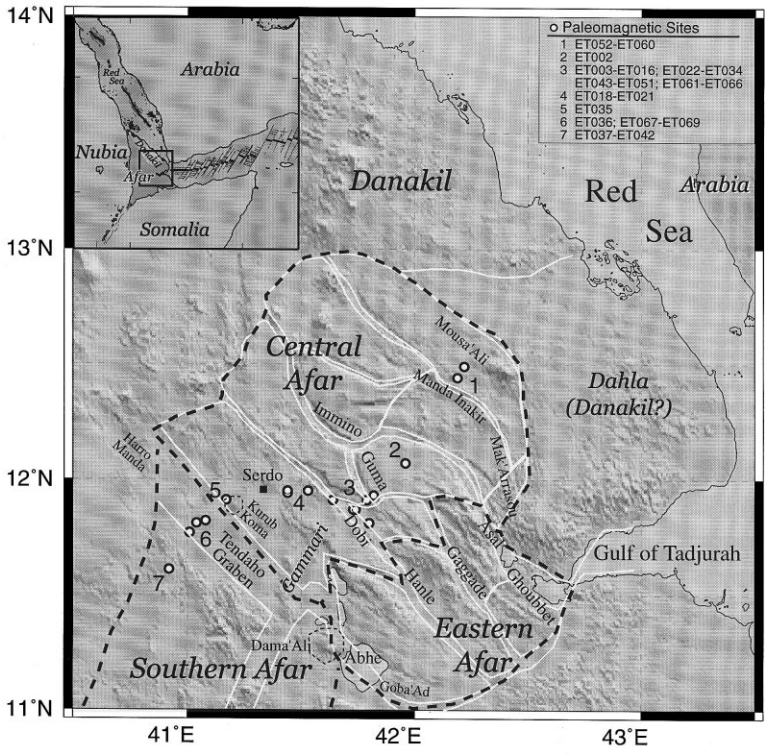


Fig. 1. Location map with paleomagnetic sample sites from this study and tectonic provinces (dashed lines). Outlines of crustal blocks used in reconstructions are shown by white lines. The topographic data are from the GTOPO3 data set provided by the United States Geological Survey. Mercator projection.

gated inland through the Gulf of Tadjurah (Fig. 1), the surrounding lithosphere evolved from a stage of diffuse extension to one of localized rifting [1–6]. In eastern Afar, deformation caused by propagation and localization has led to clockwise rotations of crustal blocks that exceed rates of 6°/myr [1,5,7–10]. Based on morphology, seismicity, geodetic measurements and volcanism, the current incipient seafloor spreading center appears to extend to at least the WNW end of Lake Asal and is propagating northwestward.

Further inland, to a region we refer to as central Afar (Fig. 1), incipient propagation and localization of extension are ongoing processes. The current focus of extension appears to be the Manda Inakir rift or nearby subrifts to the northeast (e.g., [3,6,11]), though extension appears to be more distributed across central Afar than eastern Afar. In particular, grabens in the southern part of central Afar are more seismically active than their counterparts in the southern part of eastern Afar (e.g., [1,12]) as indicated by recent earthquake swarms in the Dobi Graben [9] and near Serdo in the Tendaho Graben [13]. Furthermore, Tendaho Graben has several young volcanic edifices, including Dama'Ali, Kurub Koma, and Harro Manda (Fig. 1). Thus, at least to some extent, extensional strain is currently distributed among several rifts across central Afar, which may therefore be in an evolutionary stage analogous to what eastern Afar experienced in the Pleistocene ($\sim$1.5–0.3 Ma).

Given that rift propagation and localization appear to be responsible for the crustal block rotations observed in eastern Afar, we might expect that similar rotations are currently occurring in central Afar. To assess the rotational strain that has occurred in central Afar over the past $\sim$2 myr, we collected several hundred oriented drill cores for paleomagnetic analysis from Qua-

ternary volcanic rocks in the region between Manda Inakir and Tendaho (Fig. 1). Besides constraining the size of tectonic rotations, our paleomagnetic results provide magnetostratigraphic age constraints and insights into intriguing geomagnetic field behavior.

2. Geology

During Pliocene and Pleistocene time (~2.4–0.8 Ma), Afar was covered by a thick sequence of flood basalts, referred to as the Afar Stratoid Series (mean age 1.7 ± 0.2 Ma; Fig. 2) [2,14–17]. Subsequent faulting has produced a landscape dominated by horsts and grabens and faults and fissures, which have a dominant northwest–southeast orientation. Fault displacement is dominantly normal, with lesser amounts of strike-slip motion, which is mainly left-lateral [18,19].

The grabens are filled with detrital sedimentary units, evaporites, and volcanics [17,19]. The youngest volcanics are late Quaternary (< 1 Ma) basalt flows and spatter cones that are associated with axial volcanic ranges and volcanoes on the northeastern (Manda Inakir and Mousa'Ali) and southwestern (Harro Manda and Kurub Koma in the Tendaho Graben) sides of central Afar [3,20]. In eastern Afar, extension is focused in a single incipient seafloor spreading center, the Asal–Ghoubbet Rift [21–24], though extension and strike-slip motion had been distributed in several other subrifts (Hanle, Gaggade, and Dobi grabens) during the Pleistocene as discussed in Acton et al. [1].

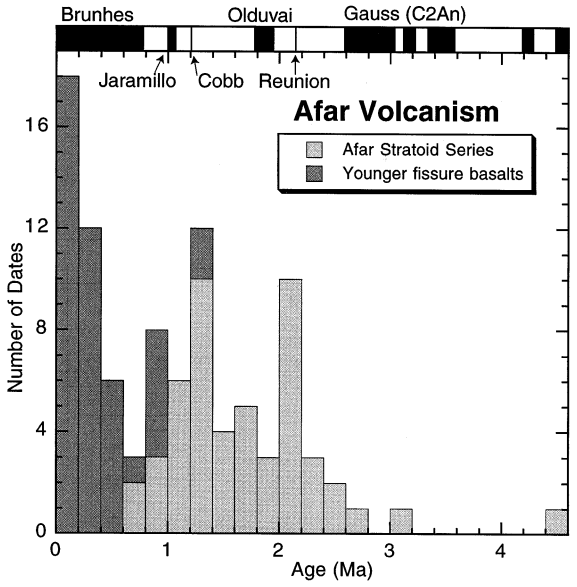


Fig. 2. Histogram of available dates for the Afar Stratoid Series and younger fissure basalts (compiled from [3,5,6,14,19–21,28,37–40]). The geomagnetic polarity time scale (top) is from Cande and Kent [41].

3. Paleomagnetic data

3.1. Sample collection

Paleomagnetic samples were collected as part of an ancillary study during a GPS campaign that was conducted in October and November 1992 [25,26]. We concentrated our sampling in the Afar Stratoid Series (62 sites), with an additional five sites in the younger volcanics of the Tendaho Rift. Thus, our paleomagnetic results are indicative only of recent tectonics, particularly of deformation that has affected the area over the past ~2 myr.

Samples, 3–8 cm long and 2.5 cm in diameter, were cored from exposures using a portable gas-powered drill and oriented with a Pomeroy orientation device that contains a magnetic and sun compass. For 383 samples (over 80%), we obtained both sun and magnetic compass orientations. Comparison shows that these typically differ by less than 2.5° and that the magnetic compass declinations are on average 1.0 ± 0.4° larger than those recorded using the sun compass. In all cases where orientations were obtained with both compasses, we use the sun compass orientation. We typically collected five to seven cores at each site, where a site is a single volcanic flow. We sampled as many individual flows as could be safely reached from thick sequences of volcanic flows that are exposed along the scarps of horst blocks. Samples at a site were generally spread 5–20 m laterally along a flow and were spaced to span roughly the entire flow vertically, with the flows typically being 0.5–7 m thick. Owing to lo-

gistical difficulties, which included hot temperatures, rugged terrain, salt marshes, and very few access roads, most sample localities were within 3 km of Highway 2 [27].

In all, we collected 444 oriented cores from 67 sites from Quaternary volcanic rocks in the region from just north of the Manda Inakir volcanic range to just south of the Tendaho–Goba'Ad graben. Six of the sites are from a horst block south of Tendaho Graben in southern Afar, five sites are from within Tendaho Graben, and the remaining 56 sites are in central Afar, north of Tendaho Graben and south of Mousa'Ali volcano.

3.2. Laboratory analysis

Laboratory analyses were conducted in the paleomagnetism laboratory at the University of New Mexico. Samples were sliced into one or two cylindrical specimens that are 2.5 cm in diameter and ~2.2 cm in height. Magnetizations of the specimens were measured in a 2G Enterprises cryogenic magnetometer (model 760R with AC SQUIDs) with an automated sample handler and an in-line alternating field (AF) unit, all of which reside in a magnetically shielded room. All specimens were progressively demagnetized using standard AF or thermal methods. Typically, each specimen was subjected to 10–15 demagnetization steps ranging from 5 to 130 mT or 120 to 680°C (data available in the **EPSL Online Background Dataset**[1], Tables A1 and A2).

3.3. Magnetic characteristics

The demagnetization results reveal fairly simple magnetization behavior as would be expected for young volcanic rocks. In general, only one or two components of magnetization are present, with the low-to-medium unblocking temperature and low coercivity component being either a small present-day overprint or an overprint related to lightning strikes (Fig. 3). These overprint directions were easily removed in all but a few samples by demagnetization up to 30 mT or 480°C. The remaining magnetization is interpreted as the characteristic remanent magnetization (ChRM) acquired as the volcanic rocks cooled shortly after extrusion. The ChRM is likely carried by pseudo-single domain titanium-poor titanomagnetite or magnetite, as evidenced by hysteresis results and the rapid decay of the majority of the magnetization between 500°C and 580°C during thermal demagnetization. In general, AF demagnetization results are less noisy than thermal demagnetization, though both methods resolve the ChRM well and give similar mean directions. For example, the mean direction computed from all sites using only thermally demagnetized samples differs from the mean using only AF-demagnetized samples by less than 0.8°.

Overall, the magnetic characteristics of the samples analyzed in this study are very similar to those described for other studies of the Afar Stratoid Series in eastern Afar [5] and near Gamarri Lake [28]. For example, the mean NRM intensity is 4.6 A/m for our samples and 4.7 A/m for those measured by Kidane et al. [28]. This value reduces to a mean of 2.8 A/m following AF demagnetization at 20 mT. The mean susceptibility is 0.023 SI volume units. Also, similar to Kidane et al. [28], we observe a small component with the same direction as the ChRM that persists up to 620–650°C in many samples (Fig. 3). In some cases, we also observe that cores taken near the top of a flow are overprinted by the overlying flow (e.g., samples ET009A and ET016A in the **EPSL Online Background Dataset**[1]).

3.4. Data analysis

The ChRM direction is estimated by fitting lines through the demagnetization data using principal component analysis (PCA). The lines were typically fitted through more than nine demagnetization steps (**EPSL Online Background Dataset**[1], Table A2). We generally did not require that the demagnetization lines pass through the origin of the demagnetization plot in the PCA analysis (the 'free' option of standard paleomagnetic PCA). For 14 of the 487 specimens ana-

[1] http://www.elsevier.nl/locate/epsl; mirror site: http://www.elsevier.com/locate/epsl

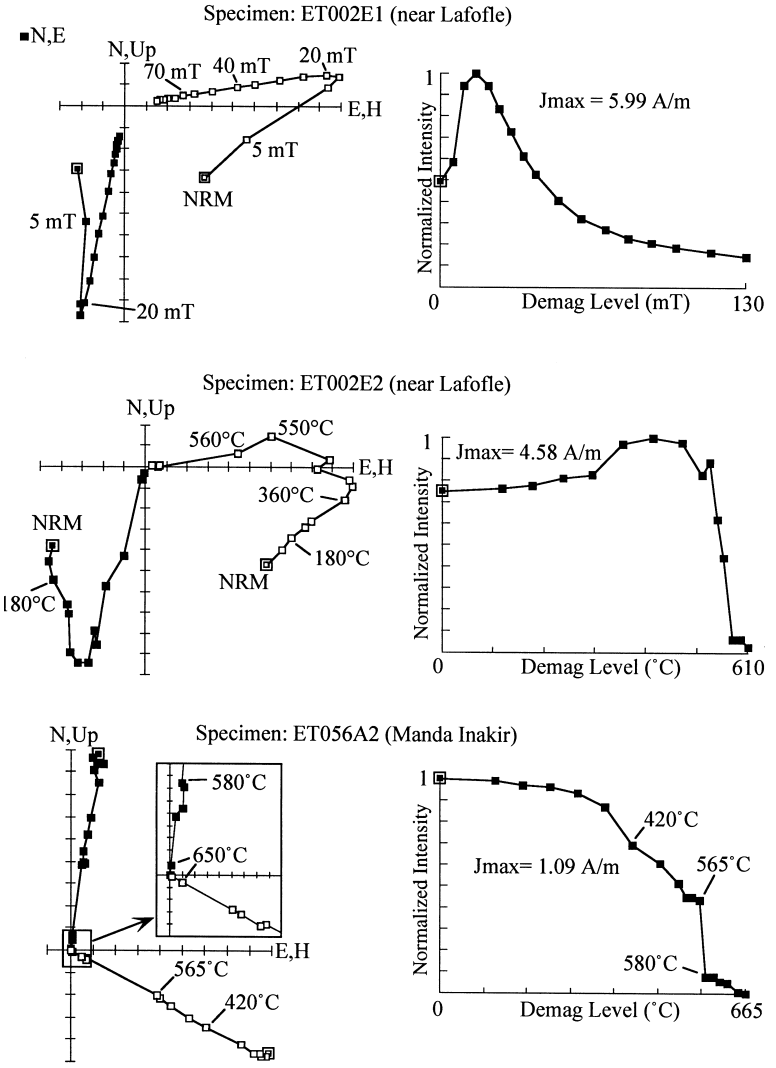


Fig. 3. Characteristic AF and thermal demagnetization results for basalt samples. Right diagrams show the intensity variation with progressive demagnetization, and left diagrams show vector end points on a vector demagnetization diagram (orthogonal projections, where open squares are inclination and solid squares are declination). AF demagnetization results for Specimen ET002E1 (top) show removal of what is likely a Brunhes-age normal-polarity overprint in peak fields of <25 mT, leaving a reversed-polarity ChRM that has a medium-to-high coercivity. Thermal demagnetization results for Specimen ET002E2 (middle) show that the magnetic mineral carrying the ChRM in this sample has an unblocking temperature concentrated between about 550°C and 580°C, though a small amount of the ChRM is present up to 610°C. The overprint is probably not completely removed until 550°C in this specimen. Thermal demagnetization results for Specimen ET056A2 (bottom), illustrating that the ChRM is also present in magnetic minerals with unblocking temperature >580°C.

lyzed, the demagnetization lines were fixed to the origin because the ChRM was only resolved in the highest few demagnetization steps. Maximum angular deviation (MAD) angles measured in the PCA were $<12°$ for all specimens used in subsequent calculations, with 80% of the specimens having MAD angles $<4°$.

Of the 444 samples (487 specimens) measured, 36 samples (39 specimens) were rejected because the ChRM direction could not be resolved well or

Table 1
Secular variation unit mean directions

Site/SV unit name	Sites combined	Site		In situ		Structural correction		Structurally corrected		*n*	*N*	*R*	*K*	α_{95}	*nX*	*NX*	*DN*	VGP (North Pole)	
		Latitude	Longitude	Inclination	Declination	Strike	Dip	Inclination	Declination									Latitude	Longitude
ET002		12.07	41.96	−10.6	191.6	0	0	−10.6	191.6	9	7	6.978	272.9	3.7	0	0	11	76.7	161.4
ET003SV	ET003–ET005	11.93	41.82	21.3	2.3	0	0	21.3	2.3	16	16	15.894	141.1	3.1	2	2	10	87.6	153.3
ET006		11.91	41.64	−23.0	183.3	N158°E	16°NE	−15.5	188.3	6	6	5.941	84.7	7.3	0	0	8	80.9	157.0
ET008		11.91	41.64	−38.2	180.4	N158°E	16°NE	−30.8	190.2	6	6	5.992	613.3	2.7	1	1	9	79.1	105.1
ET009SV	ET009–ET012	11.91	41.64	−11.7	160.1	N158°E	16°NE	−10.7	163.3	21	21	20.681	62.8	4.0	5	5	10	72.3	291.5
ET013		11.91	41.64	−13.3	173.8	N158°E	16°NE	−8.5	176.8	5	5	4.936	62.7	9.7	2	2	11	81.7	244.4
ET014SV	ET014 and ET015	11.91	41.64	45.2	343.3	N158°E	16°NE	41.6	358.2	18	17	16.423	27.7	6.9	2	2	12	77.9	33.8
ET016		11.91	41.64	6.8	351.7	N158°E	16°NE	2.8	353.0	6	6	5.982	284.6	4.0	1	1	11	77.4	255.6
ET018		11.95	41.53	−8.2	179.3	0	0	−8.2	179.3	8	8	7.984	425.9	2.7	1	1	11	82.1	226.6
ET019SV	ET019 and ET020	11.95	41.44	−20.5	180.7	0	0	−20.5	180.7	16	15	14.937	220.6	2.6	3	3	11	88.5	194.6
ET022		11.91	41.79	3.0	191.3	0	0	3.0	191.3	6	6	5.974	194.5	4.8	0	0	6	72.5	181.1
ET023@		11.91	41.79	24.2	181.9	0	0	24.2	181.9	5	4	3.982	164.0	7.2	2	2	6	65.4	217.3
ET024@		11.91	41.79	37.2	175.4	0	0	37.2	175.4	7	6	5.900	50.2	9.5	0	0	7	57.0	229.7
ET025		11.91	41.79	11.1	18.3	0	0	11.1	18.3	5	5	4.901	40.2	12.2	1	1	8	70.9	149.4
ET026SV	ET026 and ET027	11.91	41.79	39.7	22.3	0	0	39.7	22.3	12	11	10.922	128.9	4.0	1	1	9	66.2	102.2
ET028		11.91	41.79	27.9	12.4	0	0	27.9	12.4	7	6	5.976	210.4	4.6	0	0	7	77.6	116.8
ET029		11.91	41.79	1.7	359.6	0	0	1.7	359.6	5	5	4.957	92.0	8.0	1	1	9	78.9	223.9
ET030		11.91	41.79	20.5	0.4	0	0	20.5	0.4	5	5	4.971	135.8	6.6	1	1	10	88.6	205.2
ET031SV	ET031–ET033	11.91	41.79	22.1	13.7	0	0	22.1	13.7	21	18	17.767	72.9	4.1	0	0	9	76.6	132.2
ET034		11.91	41.79	17.8	4.7	0	0	17.8	4.7	6	6	5.986	353.8	3.6	0	0	9	84.6	162.5
ET035		11.91	41.17	13.4	351.8	0	0	13.4	351.8	23	13	12.978	549.0	1.8	0	0	10	80.4	279.6
ET036		11.82	41.08	9.8	359.5	0	0	9.8	359.5	9	7	6.988	503.6	2.7	10	7	10	83.1	225.2
ET037		11.61	40.92	−46.9	179.2	N170°E	3°NE	−46.3	182.3	7	6	5.879	41.2	10.6	0	0	6	73.8	48.3
ET038		11.61	40.92	−18.6	165.5	N170°E	3°NE	−18.8	166.5	5	4	3.993	433.7	4.4	2	2	9	76.6	303.8
ET039		11.61	40.92	−32.6	163.8	N170°E	3°NE	−32.9	165.7	6	6	5.940	83.0	7.4	0	0	11	74.8	337.2
ET040@		11.61	40.92	−47.3	31.8	N170°E	3°NE	−49.2	29.2	6	6	5.871	38.8	10.9	0	0	6	39.7	187.6
ET041		11.61	40.92	43.2	356.6	N170°E	3°NE	42.8	359.4	5	5	4.910	44.6	11.6	1	1	9	76.8	38.5
ET042		11.61	40.92	31.2	344.1	N170°E	3°NE	31.5	345.9	6	6	5.990	503.2	3.0	0	0	11	75.3	334.2
ET043		11.81	41.80	−38.8	190.6	N142°E	20°NE	−22.9	198.0	6	5	4.966	116.9	7.1	0	0	10	72.4	129.6
ET044		11.81	41.80	−15.8	173.1	N142°E	20°NE	−4.9	176.2	4	4	3.987	230.3	6.1	0	0	10	79.9	244.0
ET045SV	ET045 and ET046	11.81	41.80	−30.6	184.8	N142°E	20°NE	−16.2	190.9	10	10	9.987	686.4	1.8	0	0	11	78.7	149.1
ET047@		11.81	41.80	7.0	181.4	N142°E	20°NE	19.3	177.6	6	6	5.983	288.0	4.0	0	0	10	68.1	228.2
ET048		11.81	41.80	−11.9	176.2	N142°E	20°NE	−0.3	178.0	7	6	5.885	43.3	10.3	0	0	9	78.2	231.6
ET049	ET049–ET051	11.81	41.80	−22.3	165.9	N142°E	20°NE	−13.2	171.7	16	16	15.957	345.9	2.0	1	1	10	80.3	280.5
ET052		12.49	42.22	26.1	3.9	0	0	26.1	3.9	8	8	7.960	173.1	4.2	0	0	8	86.0	113.2
ET054SV	ET054–ET058	12.44	42.19	23.9	11.2	0	0	23.9	11.2	32	27	26.890	237.0	1.8	5	5	9	79.1	130.7
ET059		12.44	42.19	19.8	3.2	0	0	19.8	3.2	8	6	5.994	764.5	2.4	0	0	11	86.1	167.3
ET060		12.44	42.19	24.5	358.7	0	0	24.5	358.7	6	6	5.990	521.3	2.9	1	1	7	88.7	329.7

Table 1
Secular variation unit mean directions

Site/SV unit name	Sites combined	Site		In situ		Structural correction		Structurally corrected		n	N	R	K	α_{95}	nX	NX	DN	VGP (North Pole)	
		Latitude	Longitude	Inclination	Declination	Strike	Dip	Inclination	Declination									Latitude	Longitude
ET063		11.87	41.73	39.2	357.8	N82°E	21°NW	18.3	356.7	6	6	5.972	175.6	5.1	0	0	8	85.9	274.6
ET064		11.87	41.73	31.5	358.7	N82°E	21°NW	10.6	357.8	8	8	7.988	583.2	2.3	0	0	9	83.1	240.3
ET066		11.87	41.74	37.0	2.5	N82°E	21°NW	16.3	0.7	5	5	4.975	160.9	6.1	1	1	7	86.4	210.7
ET067		11.81	41.04	24.0	4.8	N143°E	5°NE	20.6	6.3	10	7	6.976	249.1	3.8	0	0	9	83.7	141.1
ET068		11.81	41.04	13.1	357.7	N143°E	5°NE	10.2	358.5	8	6	5.989	434.8	3.2	1	1	8	83.2	233.7
ET069		11.77	41.01	16.6	0.8	0	0	16.6	0.8	7	7	6.987	473.5	2.8	0	0	10	86.6	207.5

SV: secular variation unit comprised of samples from several flows with indistinguishable mean directions. Strike: the azimuth of the bedding strike. Dip: the angle and direction in which the bedding are inclined. *n*: number of specimens used. *N*: number of sample vectors used to calculate mean direction. *R*: resultant vector length from summing the *N* sample vectors. *K*: the precision parameter from Fisher statistical calculations. α_{95}: the 95% confidence circle for the mean direction. *nX*: number of specimens rejected. *NX*=number of samples rejected. *DN*: the average number of demagnetization steps used in the PCA to estimate the direction for each specimen from a site. @: Sites with intermediate polarity.

Fig. 4. Stereographic (Wulff) projection of secular variation unit mean directions. The solid black circle gives the expected direction for a current geocentric axial dipole (GAD) field, and the gray shaded circle gives the expected Plio–Pleistocene direction. The mean of the normal- and reversed-polarity units and their 95% confidence circles (α_{95}) are also shown, with the reversed-polarity mean also inverted for comparison. Site names are provided for the transitional or anomalous directions.

the sample direction was aberrant with respect to other samples from the same site. A sample was considered to be aberrant or an outlier if its direction was more than two angular standard deviations and $>10°$ angular distance from the site mean. In addition, we rejected six sites (ET007, ET021, ET053, ET061, ET062, and ET065) because of their large within-site directional scatter ($\alpha_{95}>20°$; full data set available as **EPSL Online Background Dataset**[1], Table A3).

The remaining 61 sites were then examined to see if they had sampled the geomagnetic field independently of stratigraphically adjacent flows. If two or more stratigraphically adjacent flows gave statistically indistinguishable directions [29], they were considered to provide a single independent measure of the geomagnetic field. Such flows were grouped into 'secular variation' (SV) units by averaging all the sample directions from the respective sites. This resulted in 44 SV units (Table 1), of which 16 have reversed polarity, 24 have

normal polarity, and four (ET023, ET024, ET040, ET047) are considered to be intermediate in polarity as their directions differ by $>35°$ from the overall SV unit mean direction (Fig. 4).

The mean of the normal site directions differs 4.7° from the antipode of the mean of the reversed site directions (Table 2, Fig. 4), suggesting that these data pass a reversal test at the B classification level of McFadden and McElhinny [30]. The fold (tilt) test is inconclusive (before and after making the tilt correction, the precision parameter $K=29.4$ and 31.8, respectively) because only 22 of the 40 unit mean directions need to be corrected for tilting, and the amount of tilting is relatively small ($<21°$ in all cases). The attitude of the volcanic flows can, however, be estimated accurately to within a few degrees because the flows are remarkably uniform in thickness over wide areas (several tens of square kilometers), suggesting that they flowed with great fluidity and that their surfaces were within a few degrees of horizontal when emplaced.

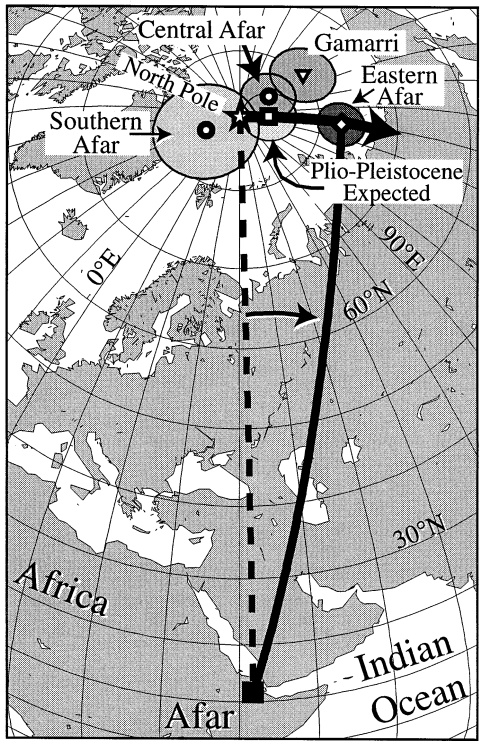


Fig. 5. Mean paleomagnetic poles are shown for central Afar (dot with darker shaded 95% confidence region), eastern Afar (diamond), Gamarri (triangle), and southern Afar (dot with lightly shaded confidence region). Also shown is the current spin axis (star) and the Plio–Pleistocene expected pole position (square). Clockwise rotation about a vertical axis within Afar results in the displacement of the poles, particularly that from eastern Afar, toward eastern Asia. Orthographic projection centered at 60°N, 40°E.

4. Results and interpretation

4.1. Tectonic implications

Mean paleomagnetic directions and poles for all the SV units and for subsets from different outcrops and regions are presented in Table 2 and Fig. 5. In general, there are too few SV units from any single outcrop or crustal block to sufficiently average geomagnetic secular variation. However, when all the SV units from central Afar are taken together, there are sufficient data to examine its motion relative to Africa and to surrounding areas.

Evidence that paleosecular variation (PSV) is adequately averaged for our central Afar sites comes from the virtual geomagnetic pole (VGP) scatter. As defined by McFadden et al. [31], the VGP scatter is the between-flow dispersion. In our case, it is the between-SV-unit dispersion relative to the mean paleomagnetic pole for central Afar, where we use a mean within-site dispersion of 6.2° and estimate the uncertainty in the VGP scatter by the bootstrap method. Based on a compilation of global paleomagnetic results from 0–5 Ma lava flows [31], the VGP scatter at the latitude of central Afar ($\sim$11.8°N) should be between 12.2° and 13.3°. When we use the 30 SV units from central Afar that have reversed or normal polarity, excluding the three SV units (ET023, ET024, and ET047) with transitional or anomalous VGPs, the VGP scatter is $10.2\pm2.4°$ (95% confidence limits). Had we used the VGPs from all 33 SV units, the VGP scatter would have been $12.2\pm3.1°$. In either case, the dispersion in the central Afar data is consistent with the global average.

To investigate possible tectonic rotations and latitudinal translations, we first compare the ob-

Table 2
Mean paleomagnetic directions and poles

Description	N	Paleomagnetic direction					Paleomagnetic Pole				
		Inc.	Dec.	R	K	α_{95}	Lat.	Long.	R	K	A_{95}
Means by polarity											
Normal	24	20.6	2.3	23.390	37.7	4.9	87.4	154.0	23.632	62.6	3.8
Reversed (inverted)	16	16.2	0.7	15.413	25.6	7.4	86.8	210.2	15.635	41.0	5.8
Normal+reversed	40	18.9	1.7	38.772	31.8	4.1	87.5	179.3	39.256	52.4	3.2
Means by outcrop											
North of Dobi Graben (ET006–ET016)	6	18.5	358.0	5.766	21.4	14.8	87.4	264.4	5.871	38.8	10.9
Near Deichooto (ET018–ET020)	2	14.4	0.0	1.988	85.8	27.3	85.4	221.6	1.997	294.8	14.6
South of Dobi Graben (ET022–ET034)	8	17.4	10.0	7.736	26.5	11.0	79.6	145.6	7.869	53.3	7.7
South of Tendaho Graben (ET037–ET042)	5	34.7	351.2	4.904	41.5	12.0	78.7	356.8	4.934	60.8	9.9
South of SE Dobi Graben (ET043–ET051)	5	11.7	2.7	4.882	33.9	13.3	83.5	195.7	4.914	46.7	11.3
North side of Manda Inakir (ET054–ET060)	3	22.8	4.4	2.988	163.1	9.7	85.5	139.2	2.988	164.6	9.6
In Dobi Graben (ET063–ET066)	3	15.1	358.4	2.994	329.9	6.8	85.5	242.6	2.997	760.9	4.5
3 km north of Dubte (ET067–ET068)	2	15.4	2.3	1.988	79.9	28.3	85.5	189.7	1.993	146.6	20.8
Means by region											
Eastern Afar[a]	71	22.4	12.6	68.023	23.5	3.5	77.3	129.3	68.801	31.8	3.0
Gamarri Lake[b]	27	10.1	8.1		47.8	4.1	79.7	170.2			4.1
Central Afar	30	17.0	3.6	29.196	36.1	4.4	85.3	171.1	29.517	60.0	3.4
Southern Afar	10	24.4	355.6	9.698	29.8	9.0	85.4	332.9	9.834	54.3	6.6

Eastern Afar: the region between Goba'Ad Graben and Asal–Ghoubbet Rift. Gamarri Lake: a section sampled by Kidane et al. [28] near the boundary between eastern and central Afar, north of Gammari Lake. Central Afar: the region between Tendaho Graben and Mousa'Ali Volcano sampled in this study. Southern Afar: the region within and south of Tendaho Graben.
Outcrop means are given only for outcrops where more than one SV unit was sampled.
N: number of secular variation units used to calculate the mean. These have been corrected for observed tilting. R: resultant vector length from summing the N vectors (directions or poles). K: the precision parameter from Fisher statistical calculations. α_{95} and A_{95}: the 95% confidence circle for the mean direction and mean pole, respectively.
[a]Estimates from table 1 of Acton et al. [1] based on paleomagnetic data from Galibert et al. [46] and Courtillot et al. [5].
[b]Estimates from table 3 of Kidane et al. [28].

served directions with the expected present-day geocentric axial dipole (GAD) direction and with the expected Pliocene–Pleistocene direction for Africa (Table 3). Given that the paleomagnetic field is represented by a GAD and that lithospheric plates move slow relative to Earth's spin axis, we would expect the ~2 Ma reference poles from stable portions of Africa to differ insignificantly from Earth's spin axis. However, the reference paleomagnetic pole for the Plio–Pleistocene (86.4°N, 132.7°E with $A_{95} = 3.4°$; computed from seven African poles with mean ages 1–4 Ma given in table 1a of Besse and Courtillot [32]) differs from the spin axis, though only marginally (Figs. 4 and 5). This pole gives an expected direction with an inclination of 22.5° and a declination of 3.7° in Afar, resulting in tectonic rotation estimates that are 3.7° less than if the expected GAD direction is used. In the following, we quote rotations relative to the expected GAD direction, though rotations relative to either expected direction are given in Table 3. We assume a 3° uncertainty ($A_{95} = 3°$) for the GAD to account for the uncertainty in its position over the past 2 myr caused by plate motions and small nondipole field components.

Comparison of the paleomagnetic direction from central Afar with either expected direction

indicates no statistically significant rotations have occurred, though rotations as large as 8° are permitted (rotation = 3.6 ± 4.4°). As observed in other studies of the Afar Stratoid Series, the inclination is marginally shallow when compared with the expected GAD direction (flattening = 5.7 ± 5.5°). This could result from a number of phenomena, of which the most likely are: (1) a paleomagnetic field that contains a small geocentric axial quadrupole component (e.g., [33,34]), (2) small, unremoved secondary overprints, (3) tectonic tilt corrections that fail to account for southward-tilting basalt flows or that overcorrect for original deposition of flows that dip to the north, and (4) inadequate averaging of geomagnetic secular variation. Given the relatively large number of SV units and the small tilts, the latter two are unlikely. A geomagnetic field that has a 5% quadrupole, on the other hand, is observed in most Quaternary paleomagnetic data sets and would reduce the flattening to an insignificant 1.9 ± 5.6°.

A smaller set of data from southern Afar indicated no significant rotations (−4.4 ± 8.3°) or translation (−2.1 ± 8.4°; Fig. 5). In this case, however, there are too few SV units to average secular variation sufficiently.

We next examine the motion of central Afar, eastern Afar, and the Gamarri Lake area relative to each other (Table 3). Eastern Afar has a statistically significant clockwise rotation of ∼9° relative to central Afar and ∼4.5° relative to Gamarri. Similarly, Gamarri has rotated ∼4.5° clockwise relative to central Afar. Based on topography and structural features, Gamarri is likely a part of eastern Afar (Fig. 1) or possibly a small crustal fragment caught in a transfer zone between central and eastern Afar. In either case, the observed ∼8° of clockwise rotation for Ga-

Table 3
Vertical-axis rotations and inclination flattening estimates

Description (rt. = relative to)	Observed direction			Expected direction			Rotation		Flattening	
	Inc.	Dec.	α_{95}	Inc.	Dec.	A_{95}	R	ΔR	F	ΔF
Central Afar rt. present-day GAD direction	17.0	3.6	4.4	22.7	0.0	3.0	3.6	4.4	5.7	5.5
Central Afar rt. Plio–Pleistocene expected direction	17.0	3.6	4.4	22.5	3.7	3.4	−0.1	4.6	5.5	6.0
Eastern Afar rt. present-day GAD direction[a]	22.4	12.6	3.5	22.1	0.0	3.0	12.6	3.9	−0.3	5.1
Eastern Afar rt. Plio–Pleistocene expected direction	22.4	12.6	3.5	21.9	3.7	3.4	8.9	4.1	−0.5	5.6
Gamarri rt. present-day GAD direction[a]	10.1	8.1	4.1	22.3	0.0	3.0	8.1	4.1	12.2	5.4
Gamarri rt. Plio–Pleistocene expected direction	10.1	8.1	4.1	22.2	3.7	3.4	4.4	4.3	12.1	5.9
	Inc.	Dec.	α_{95}	Inc.	Dec.	α_{95}	R	ΔR	F	ΔF
Gamarri rt. central Afar	10.1	8.1	4.1	17.0	3.6	4.4[b]	4.5	3.3	6.9	4.8
Eastern Afar rt. Gamarri	22.4	12.6	3.5	10.1	8.1	4.1[b]	4.5	3.8	−12.3	4.3
Eastern Afar rt. central Afar	22.4	12.6	3.5	17.0	3.6	4.4[b]	9.0	3.0	−5.4	4.5

Central Afar: the region between Tendaho Graben and the Manda Inakir volcanic range sampled in this study. Eastern Afar: the region sampled by Courtillot et al. [5] between Goba'Ad Graben and the Asal–Ghoubbet rift zone. Gamarri: a section sampled by Kidane et al. [28] between eastern and central Afar, north of Gammari Lake.

Mean site locations: 11.8°N, 41.7°E for Central Afar; 11.4°N, 42.1°E for Eastern Afar; and 11.6°N, 41.7°E for Gamarri.

Present-day GAD: expected direction based on the current geocentric axial dipole (assumed uncertainty 3°, see text).

α_{95}: the 95% confidence circle for the mean direction. A_{95}: The 95% confidence circle for the mean paleomagnetic pole. $R \pm \Delta R$: vertical axis rotation and its 95% confidence limits [47,48]. Positive values indicate clockwise rotations. $F \pm \Delta F$: inclination flattening and its 95% confidence limits. Positive values indicate shallower than expected inclinations.

[a]Estimates from table 1 of Acton et al. [1] based on paleomagnetic data from Galibert et al. [46] and Courtillot et al. [5].

[b]These are α_{95} values.

marri is consistent with expectations given the paleomagnetic uncertainties.

We envision two scenarios to explain the observed rotations. In both, rift propagation and localization is nearly complete in eastern Afar and has resulted in clockwise rotations that average about 11° over that region. In the first scenario, similar processes are under way in central Afar, and have resulted in a small (~4°), possibly negligible, clockwise rotation (Fig. 6). In this scenario, clockwise rotations will increase over time until stable seafloor spreading has been established presumably on the NNE side of central Afar. In the second scenario, small or insignificant rotations may be all that will occur in central Afar if rift propagation and localization have occurred rapidly across the area rather than gradually as in eastern Afar. In this scenario, extension should already be focused on the NNE side of central Afar rather than the graben to the southwest. Seismic events in Tendaho (Serdo) and Dobi Grabens argue against this. In either case, it is clear that extension across Afar has been inhomogeneous, leading to widespread and intense crustal fragmentation. Extension has also progressively migrated to rifts on the northeastern side of the region and these rifts have generally opened progressively toward the northwest, two phenomena that ultimately caused the clockwise rotations.

Though overly simplistic, the crustal fragments of central Afar can be reconstructed similar to those of eastern Afar (e.g., [1,7,8]). The model we develop is consistent with the key constraints, which are the motions of the bounding plates, the paleomagnetic data, the ages of geologic units, and the geometry of grabens, horsts, and transfer zones. Here we use recently revised estimates for the relative motions between Nubia, Somalia, Arabia, and Danakil [35,36]. For the interior of Afar, we begin by dividing the region into a relatively small number of semi-rigid crustal blocks, which are defined by bounding grabens or highly faulted zones (Fig. 1). Like a puzzle, the pieces can be roughly reconfigured to their pre-rift geometry by simply closing the gaps (grabens) between the blocks while simultaneously restoring the past 1.8 myr of motion between the bounding plates. Again, this is an oversimplification: each of the blocks has deformed somewhat, however, their internal deformation is small relative to the large amount of deformation that has occurred at their boundaries. Similarly, the extension is large in the grabens, but not infinite. In addition, each block has likely had a more complex rotation his-

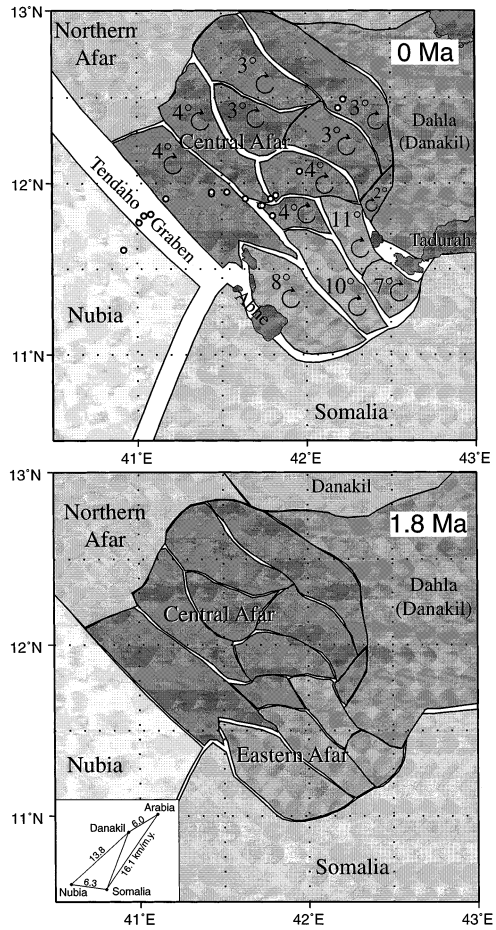


Fig. 6. Reconstruction of crustal blocks from 1.8 Ma to the present. The rotation each block sustains is given next to the arrow, which shows the sense of rotation. The velocity diagram in the lower left corner gives the relative velocities between the bounding plates for a point at 12°N, 42°E based on Chu and Gordon's [35,36] rotation poles. The crustal blocks are the same as those shown in Fig. 1. Faulting and volcanism in the southernmost Danakil block, which we define as the Dahla block, indicate that the greater Danakil block may be deforming. The relative motion between Danakil and Dahla is probably small, however, so we treat Dahla as part of a rigid Danakil block. Paleomagnetic sites are shown by hollow circles.

tory than indicated by the mean paleomagnetic results for a region comprised of multiple blocks. Ultimately, the crustal fragments are likely to have sustained small differential rotations, perhaps similar to those shown in Fig. 6, as they interact with each other while riding in a diffusely, though inhomogeneously, extending mantle.

4.2. Magnetostratigraphic observations

Based on the radiometric ages, the geomagnetic field would have had reversed polarity during roughly 83% of the 0.8–2.4 Ma interval in which the Afar Stratoid Series was extruded. However, only 36% of the SV units sampled have reversed polarity. The large percentage of normal-polarity SV units (55%) could be a statistical fluke, but probably suggests instead rapid extrusion in a narrower time interval. The mean age of all radiometric dates from the Afar Stratoid Series is 1.7 ± 0.2 Ma [3,5,6,14,19–21,37–40], which spans much of the Olduvai Chron (Chron 2n; 1.77–1.95 Ma on the time scale of Cande and Kent [41]). It seems likely that much of the Afar Stratoid Series was extruded during this subchron, though extrusion should include an interval beginning prior to the Réunion event (Chron 2r.1n; 2.14–2.15 Ma), as indicated by the results of Kidane et al. [28], and extending into the Matuyama Chron (0.78–1.77 Ma) (Fig. 2).

4.3. Geomagnetic observations

4.3.1. Preferred transitional geomagnetic field states and rapid polarity reversals

As discussed above, we excluded four (ET023, ET024, ET040, and ET047) of the 44 SV-unit mean directions because they were further than 35° from the mean direction of all the SV units (Fig. 4). When converted to VGPs, two of these (ET024 and ET040) are more than 35° from Earth's spin axis, whereas the other 42 SV units are within 25° of the spin axis. Site ET024 is from a sequence of lava flows sampled just north of Dobi Graben, and Site ET040 is from a sequence of lava flows sampled south of Tendaho Graben (Fig. 7). In both cases, the aberrant VGPs are from a single flow that is underlain by normal-

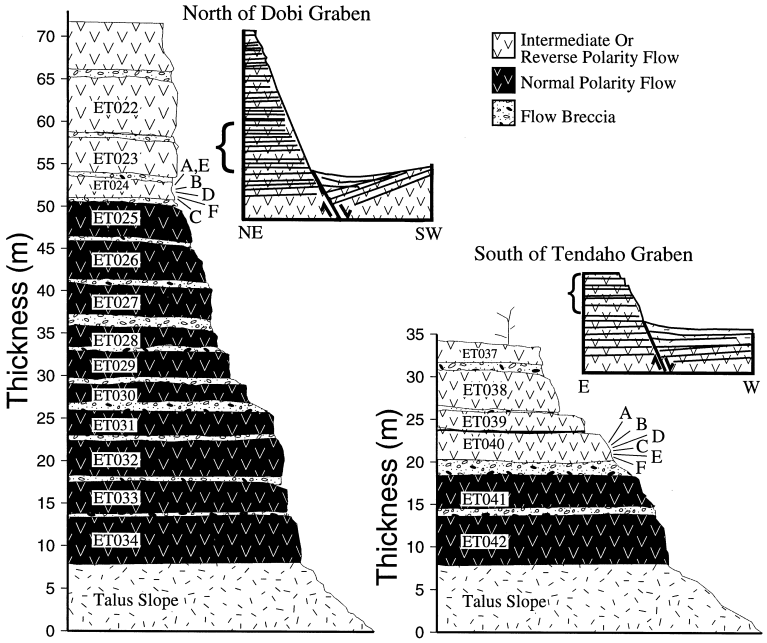


Fig. 7. Schematic cross-sections of two lava sequences that contain transitional lava flows (ET024 and ET040). The relative sample locations in the transitional flows are give by letters A–E, where A refers to Sample ET024A or ET040A.

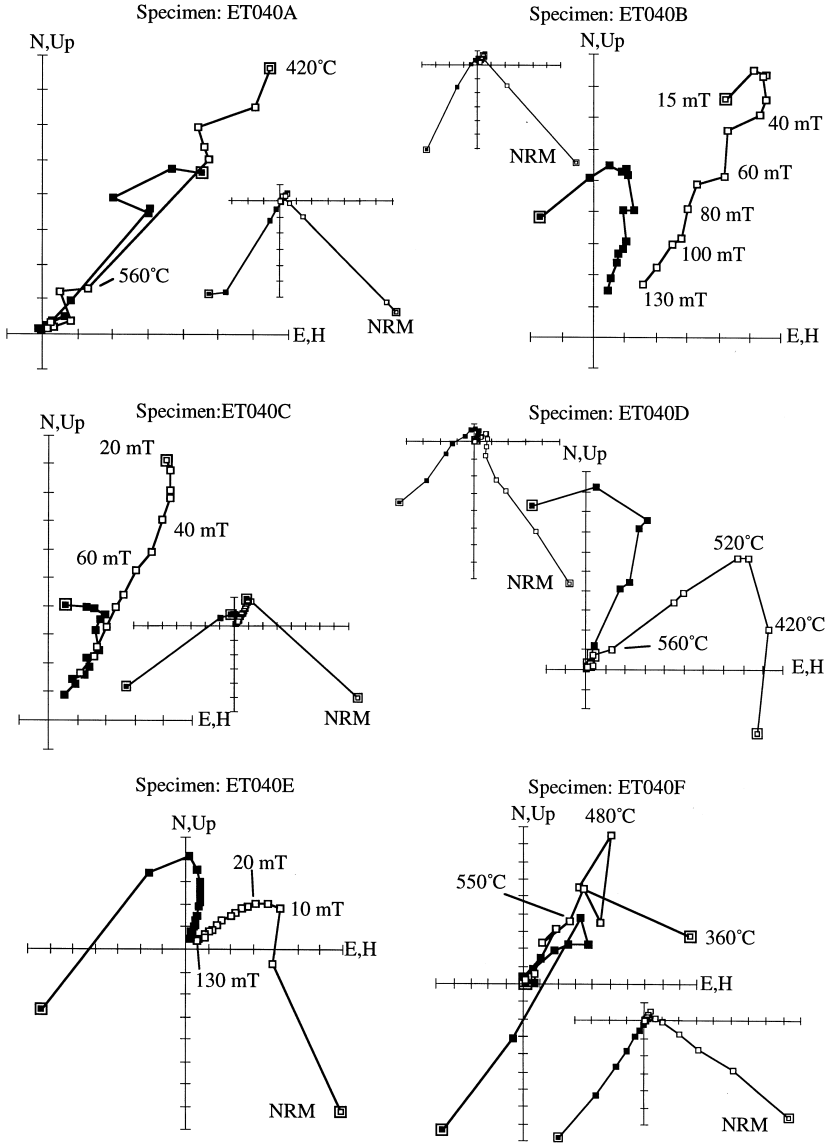


Fig. 8. AF and thermal demagnetization results for the six basalt samples collected from a lava flow extruded during a polarity transition. The small insets show the initial NRM direction, which has a positive inclination and a southwesterly declination. Removal of this low-coercivity/low-to-medium-unblocking-temperature component reveals a component with a negative inclination and northeasterly declination.

polarity flows and overlain by reversed-polarity flows. Hence, these are true transitional VGPs rather than excursional or anomalous VGPs.

A third transitional VGP is recorded by a secondary component, which is present in all six samples from Site ET040. This component has a low coercivity/low-to-medium unblocking temperature, which is removed by about 30 mT AF or 500°C thermal demagnetization (Fig. 8). The mean direction of this component has an inclination of 43.2°, a declination of 217.5°, and an α_{95} of 7.6°, which is significantly different from the

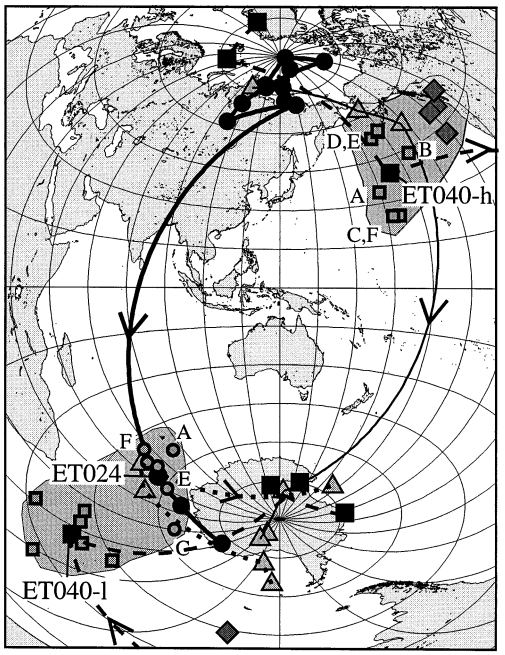


Fig. 9. Transitional VGP paths for the lava sequences from north of Dobi Graben (thick solid line connecting black circles), south of Tendaho (dashed line connecting black squares), and Gamarri Lake region (thin solid line and dotted line connecting shaded triangles). VGPs for eastern Afar sites (diamonds), and for samples from Sites ET024 (small gray circles) and ET040 (small gray squares) are also shown. The antipodal concentrations of transitional VGPs are shown by shaded patches.

overlying lava flow. Thus, reheating and partial remagnetization by the overlying flow cannot explain this component. Instead we interpret both components as transitional directions, with the high-coercivity/high-unblocking-temperature component (referred to as ET040-h) representing the field direction recorded as the flow was cooling below the Curie temperature of magnetite (580°C) and the other component (ET040-l) representing a new transitional field direction that was recorded after the flow cooled below about 500°C.

The transitional VGPs for ET040-h and ET040-l are 50.3° from the North Pole and 51.8° from the South Pole, respectively (Fig. 9). The transitional VGP from Site ET024 is 35.4° from the South Pole. Site ET023, which overlies ET024, has a VGP that is 23.0° from the South Pole and is part of a transitional VGP path that goes from the North Pole to the South Pole (Fig. 9).

When these transitional VGPs are plotted with other transitional and anomalous VGPs from around Afar, they appear to be concentrated in two, nearly antipodal regions (Fig. 9). The other transitional or anomalous VGPs come from a sequence of lava flows sampled near Gamarri Lake, which record one or possibly both of the Réunion events [28], and from sites distributed across eastern Afar [5]. In ascending order, the Gamarri section contains 19 reversed-polarity lava flows, seven normal-polarity flows, two transitional flows, and four reversed-polarity flows. The lower reversed-polarity interval contains an interval of anomalously high PSV, which was recorded by two flows (GA06 and GA07). These VGPs plot nearly on top of the VGPs from Sites ET023–ET024 (Fig. 9). The transitional VGP paths cannot, however, represent the same time interval because the VGPs progress from north to south on one path during a reversal and from south to north on the other path during an interval of high PSV. The two transitional VGPs from the upper reversal at Gamarri (GB18 and GB19) fall in the same region as Site ET040-h. In this case, these could represent the same normal-to-reverse transitional path. Furthermore, four transitional or anomalous VGPs from eastern Afar (Sites B7, B12, C5, and C13 of [5]) fall within the northern Pacific region, though a fifth site (B18) does not fall in either of the regions with high VGP concentrations.

Similar antipodal concentrations of transitional/excursional VGPs have been noted by Hoffman [42], though neither of the regions from our study corresponds exactly to the 'cluster' regions designated by Hoffman. Our southern hemisphere concentration does, however, overlap that of Hoffman, and would lie roughly on top of a VGP cluster from the Oligocene transitional VGP path recorded by the Liverpool Volcano in Australia (see figures 1 and 5a of [42]). These concentrations may provide new examples of the quasi-stationary geomagnetic field configurations during transitional and excursional events, though such an observation is very speculative given the limited number of transitional VGPs from Afar.

Our results from Site ET040 support Hoffman's [42] hypothesis that the field reverses rapidly between longer-lived (possibly inclined dipolar) field configurations. For the Site ET040 lava flow, which is ~4.5 m thick, to have recorded nearly antipodal transitional components above and below ~500°C would indicate that the jump from a northern-hemisphere transitional state to a southern-hemisphere one occurred in less than a few weeks. Directional changes during the jump would have occurred at rates comparable to or faster than the 6°/day estimated by Coe et al. [43] for a flow from Steens Mountain, Oregon.

Site ET024 may also record a rapid field transition. The sample VGPs from this site display elongation along the transitional VGP path. This may happen by random chance, or it could have resulted from the geomagnetic field changing rapidly during the time it took the flow to cool through the spectra of magnetic blocking temperatures, which should have been only a few days for the ~0.6-m-thick flow (e.g., [43]). Alternatively, the overlying flow could have partially remagnetized the underlying transitional flow, with possible accompanying low-temperature oxidation (e.g., [44]). The distribution of the six samples and their VGPs (Figs. 7 and 9) does not provide clear support for either of these possibilities. Ideally this flow should be resampled at much higher resolution to examine the possibility that it has recorded rapid field changes.

5. Conclusions

Several notable conclusions can be drawn from the Afar paleomagnetic data. First, the abundance of normal-polarity lava flows, along with radiometric ages, indicates that the Afar Stratoid Series was probably extruded over a fairly narrow interval focused near the time of the Olduvai chron (1.77–1.95 Ma), though beginning prior to the Réunion event (2.14–2.15 Ma). Second, transitional or anomalous VGPs appear to be concentrated in two, nearly antipodal regions. Third, antipodal transitional directions recorded by a ~4.5-m-thick flow (Site ET040) as it cooled indicate that directional changes during a reversal can be astonishingly fast, possibly occurring as a nearly instantaneous jump from one inclined dipolar state to another in the opposite hemisphere. A less impressive example of rapid directional changes occurring while a lava flow cools may be evidenced by the elongation of sample VGPs along a transitional VGP path for Site ET024, though this flow could also have been partially remagnetized by overlying flows. Fourth, vertical-axis rotations within central Afar are less than about 8°, and are negligible within the uncertainties. Furthermore, paleomagnetic comparison indicates that eastern and central Afar have statistically different rotation histories, with eastern Afar having sustained $9.0 \pm 3.0°$ more clockwise rotation than central Afar.

We attribute the smaller rotations in central Afar to two main factors. First, propagation and localization within central Afar are still in progress, whereas the process is nearly completed in eastern Afar. Thus, more rotation may occur as extension distributed across central Afar becomes focused into a single seafloor spreading center. Second, the rate at which propagation and localization are occurring in central Afar may be more rapid than in eastern Afar, which would result in net rotations for central Afar that are smaller than eastern Afar. The propagation and localization rates, along with the overall rate of opening between the major plates, are critical factors in determining the amount of rotation that can occur as illustrated for oceanic microplates by Engeln et al. [45]. Geodetic studies, such as the GPS network initiated by Jackson et al. [26], combined with paleomagnetic studies of rotations like this one and other geologic investigations, will ultimately allow the determination of the rate of propagation and localization throughout Afar as stable seafloor spreading is established.

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